



DR USEN UBOH PH.D

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EXECUTIVE SUMMARY

Innovative Data and Solution Architect with 15+ years of experience delivering enterprise-scale data and integration solutions across healthcare, SaaS, and highly regulated industries. Expert in aligning data architecture, integration architecture, and MDM strategy with enterprise transformation initiatives across Azure, AWS, and GCP cloud ecosystems. Proven success designing scalable data architectures leveraging Azure Databricks, Synapse, and Data Lake, alongside event-driven, API-based integration frameworks connecting Salesforce, SAP, and on-prem systems.

Specializing in Master Data Management (MDM) with deep hands-on experience in Reltio, Informatica MDM, and SAP MDG, complemented by strong data governance expertise using One Trust, Alation, and Collibra. Notable achievements include spearheading global MDM research at Gartner, architecting domain-wide master data and integration solutions at Blue Cross and Humana, and delivering real-time, cross-cloud data pipelines that enable governed analytics and AI/ML innovation. Committed to building secure, compliant, and business-aligned data ecosystems that drive measurable enterprise value.

CORE SKILLS & TECHNOLOGIES

Solution Architecture Focus Areas:

Integration & API Architecture (REST, Event-Driven, SOA)
Enterprise Application Modernization
Cross-System Data & Process Orchestration
Solution Blueprints, Reference Architectures & Transition Roadmaps

MDM & Data Management Platforms:

Reltio, Informatica MDM, SAP MDG, Semarchy xDM, Stibo STEP, Tamr, Profisee, Atacama

Data Governance, Privacy & Compliance:

OneTrust, Collibra, Alation, BigID, TrustArc, Microsoft Purview, HIPAA, GDPR, CCPA

Cloud & Data Platforms:

Microsoft Azure (Databricks, Synapse, Data Lake, Purview, ADF),
AWS (Glue, Redshift, S3, Lambda),
Google Cloud Platform (Big Query, Dataflow, Pub/Sub),
Snowflake, Salesforce

Integration & Data Engineering:

Azure Data Factory (ADF), Databricks Delta Live Tables, Apache Spark,
ETL/ELT Design, API Management (Azure API Gateway, Mulesoft),
SOAP/REST Services, Event-Driven Architectures (Kafka, Event Hubs),
SAP SOA Integrations

Programming Languages & Tools:

SQL, Python, PySpark, JSON, Jupyter, Eclipse, Visual Studio, pgAdmin, GitHub, PowerShell

Architecture & Design Expertise:

Enterprise Architecture, Solution Architecture, Integration Architecture,
Data Architecture, Data Modeling (Conceptual/Logical/Physical), Metadata Management,
Reference Architecture Design, Cloud Cost Optimization

Industries:

Healthcare, Life Sciences, Retail, Public Sector, Private Sector, Financial Services, E-commerce, Supply Chain, Aesthetics, Constructional Engineering

EXPERIENCE**Enterprise Data Architect / Micheal Baker International (UAM Consulting)***05/2024 - Present*

- Led enterprise-scale data modernization and governance transformation across 30+ divisions and 90+ offices, unifying fragmented engineering, geospatial, and environmental datasets through a cloud-native, metadata-driven data fabric supporting resilient infrastructure, digital twin enablement, and full asset lifecycle analytics.
- Architected a 10+ TB scalable Azure ecosystem using Databricks (Delta Lake), Synapse Analytics, Azure Data Factory, and Python/Spark, establishing bronze–silver–gold data layers, event-driven APIs, and CI/CD automation (GitHub Actions, Terraform, ARM). Reduced environment setup effort by 60% and accelerated release cycles from 3 weeks to 5 days through infrastructure-as-code and automated deployment patterns.
- Improved analytics delivery by 40%, data reuse by 52%, and accuracy by 33%, cutting reconciliation errors and manual data prep by 35%—resulting in \$2.1M annual savings and faster asset-renewal decisions for 1,000+ stakeholder users across Azure, AWS, and hybrid cloud platforms.
- Developed digital twin–aligned conceptual, logical, and physical data models supporting \$15B+ in public infrastructure assets, standardizing master, reference, spatial, and transactional data domains. Led MDM initiatives using Reltio / Informatica / SAP MDG, designing survivorship rules, entity resolution, and stewardship workflows for asset, location, vendor, and project master data.
- Implemented event-driven and service-based integration frameworks (REST, SOAP, Azure Event Hubs, message queues) to synchronize ERP, GIS, asset management, and external partner systems in real time, eliminating engineering file drops and decoupling application dependencies.
- Established an enterprise metadata and governance framework (TOGAF, DAMA-DMBOK) using Collibra, Alation, and OneTrust—cataloging 10,000+ datasets, enabling lineage tracking, enforcing policy controls, and reducing audit exceptions by 70% while improving traceability by 95%.
- Designed high-performance SQL and Snowflake analytical marts optimized for spatial joins, time-series models, and multi-dimensional reporting, powering self-service analytics for compliance, environmental monitoring, and infrastructure planning.
- Launched enterprise data-quality dashboards (150+ KPIs) integrated with stewardship SLAs, triggering automated workflows to surface anomalies and enforce accountability across project teams, positioning the firm as a leader in smart infrastructure, sustainability analytics, and predictive maintenance.
- Collaborated with cloud and DevOps teams to operationalize CI/CD pipelines, enforce coding and modeling standards, and mentor 15+ junior engineers and analysts on cloud-native data engineering, data architecture, and governance best practices.

Data and Martech Architect / AbbVie (UAM Consulting)*04/2024 - Present*

- Architected enterprise Martech and data-integration modernization, unifying customer, behavioral, and product data across AbbVie's global digital ecosystem. Designed a privacy-compliant, real-time customer data platform (CDP) integrating Segment, Reltio MDM, and PIM systems to enable AI-driven personalization and omni-channel activation across 50+ brands and web/mobile properties, solving multi-system fragmentation and identity drift.
- Reduced event loss by 25%, eliminated 35% duplicate profiles through Reltio-aligned identity resolution, and improved targeting accuracy by 20%. Increased campaign ROI and reduced time-to-activate audience segments by 40% across Meta, Google, TikTok, Pinterest, Salesforce Marketing Cloud, and Snowflake destinations.
- Architected a global real-time data fabric integrating Segment CDP, Reltio MDM, Azure Data Factory, Databricks, Synapse, and Snowflake—processing hundreds of millions of digital events monthly. Built bi-directional pipelines and enrichment workflows combining event streams with golden customer/product records using hashed identifiers, trait mappings, and survivorship rules.
- Designed and deployed Segment instrumentation across web, mobile, and backend apps using JavaScript SDKs, HTTP API, and cloud functions; configured governed Destinations and Functions for compliant activation to ad-tech and mar-tech platforms, enabling first-party audience portability without violating consent policies.
- Implemented Segment Protocols for schema validation, naming standards, and event governance, embedding CI/CD validations into deployment workflows to enforce alignment across all digital properties and global markets.

- Built real-time monitoring dashboards using Azure Monitor, Log Analytics, Power BI, and Reltio Console to track activation health, drop rates, schema drift, and MDM sync failures, improving observability and reducing remediation cycles.
- Defined enterprise taxonomy, identity attributes, and privacy classification frameworks aligned to GDPR/CCPA; embedded consent tagging, lineage tracking, and automated compliance scoring into CDP + MDM pipelines, achieving 98% compliance and reducing manual audit overhead.
- Configured Reltio match/merge logic, business hierarchies, and API-based synchronization to legacy ERP/CRM systems, improving product and customer domain governance and enabling unified measurement across paid, owned, and clinical-engagement channels.
- Directed cross-functional Agile teams (12 engineers, data stewards, analysts) to deliver high-velocity integrations; authored AbbVie's global MarTech Architecture Blueprint for identity unification and AI-driven optimization—now powering predictive analytics, retention scoring, and dynamic personalization at scale.
- Performed CAPEX cost-benefit analysis across CDP, MDM, and analytics workloads, generating leadership reporting on ROI, adoption, and spend optimization for ongoing Martech investment strategy.

Enterprise Data Architect / Campbell Soup Company (UAM Consulting)

01/2023 – 06/2025

- Led the global Master Data Management (MDM) transformation across snacks, meals, and beverages divisions, establishing a single source of truth for product, supplier, and customer data that unified fragmented systems, strengthened governance, and improved data-driven decision-making across 1,000+ downstream systems and users.
- Reduced duplicate records by 40%, improved reporting accuracy by 30%, and cut supplier/product onboarding time by 50%, saving hundreds of operational hours annually while enhancing agility and time-to-market.
- Architected enterprise-grade MDM platforms (Semarchy xDM, Reltio) integrated with Azure Data Factory, Event Grid, Service Bus, and Snowflake/Synapse, processing millions of records with real-time synchronization across SAP S/4HANA, CRM, and commerce systems.
- Built match/merge and survivorship logic, low-latency ingestion pipelines, and API-driven services, enabling high-quality master data and rapid data access for supply-chain, marketing, and analytics domains.
- Implemented enterprise governance and stewardship frameworks (Collibra, Alation) with automated quality validation and anomaly detection, increasing data reliability by 45% and ensuring audit-readiness for food-safety and regulatory traceability.
- Defined metadata standards, KPI dashboards, SLA metrics, and stewardship workflows under Agile delivery, embedding governance and quality control into each sprint cycle; created a reusable MDM blueprint library now serving as a foundation for future domains and global expansion.

Enterprise Data Architect / Blue Cross Blue Shield MA (UAM Consulting)

03/2023 – 04/2024

- Led enterprise data architecture and MDM modernization across Consumer, Member, and Provider domains, unifying fragmented data from claims, enrollment, CRM, and analytics systems to enable value-based care, regulatory compliance, and advanced analytics across millions of members and providers.
- Reduced duplicate records by 40%, improved data-quality scores by 25%, and shortened member lifecycle operations by 30%, strengthening accuracy, onboarding speed, and downstream data usability across core health-plan platforms.
- Architected and deployed scalable master-data infrastructure integrating Reltio MDM, Azure Data Factory, Databricks, Synapse, Snowflake, and Salesforce—supporting real-time APIs, near-real-time pipelines, and batch synchronization across operational, transactional, and analytical systems.
- Designed Reltio entity structures, hierarchies, match/merge/survivorship rules, and performance-optimized golden-record logic, improving identity resolution efficiency and API throughput for high-volume member and provider datasets.
- Developed end-to-end master data flow from Reltio to Azure Data Lake (ADLS) using Databricks bronze-silver-gold architecture, enabling curated analytical layers for population health, actuarial modeling, and provider network intelligence.
- Established enterprise metadata and lineage tracking through Collibra and Alation, cataloging master, reference, and transactional domains; automated stewardship workflows achieved 100% traceability and improved audit readiness.

- Defined and enforced enterprise data-governance architecture (TOGAF, DAMA-DMBOK) with HIPAA/GDPR-aligned classification, retention, and access-control models; automated compliance validation reduced audit prep time by 40% and strengthened security posture.
- Partnered with data governance, enterprise architecture, and integration teams to operationalize standards for data modeling, lineage, and stewardship accountability, increasing trust and reuse of master data across actuarial, care-management, and member-experience programs.
- Optimized Reltio platform performance through tuning, rule-refinement, and API load scaling, supporting millions of daily transactions and eliminating latency driven by match-merge processes.
- Delivered a trusted, compliant, and analytics-ready master-data foundation improving provider accuracy by 20% and positioning BCBS MA for AI-enabled analytics, claims optimization, and regulatory modernization initiatives.

Lead Data Architect / Humana Healthcare (UAM Consulting)

06/2022 - 03/2023

- Delivered a secure, scalable, and governed enterprise data layer enabling real-time care coordination, regulatory reporting, and predictive analytics, positioning Humana for AI-driven healthcare insights and value-based care expansion.
- Led enterprise-scale MDM design and deployment across Consumer, Member, and Provider domains using the Reltio Enterprise SaaS platform, integrating Azure, Snowflake, Salesforce, and OneTrust to deliver compliant, trusted, and analytics-ready master data used across operational, clinical, and engagement systems.
- Improved data-trust scores by 30%, reduced duplicate records by 50%, and increased query performance by 25% by optimizing match/merge logic, survivorship rules, and Reltio performance tuning for high-volume member and provider workloads.
- Defined Reltio entity models, hierarchies, match/merge rules, survivorship logic, and workflow automation aligned with enterprise MDM methodology; implemented search-before-create validation and API-based data mappings to Salesforce to prevent duplication and enforce golden-record consistency.
- Architected and developed high-volume ingestion and reconciliation pipelines using Azure Data Factory, APIs, and Reltio Connectors to synchronize master data across CRM, provider, claims, and external partner systems in real time.
- Embedded HIPAA, PHI, and GDPR compliance controls using OneTrust for privacy management, implementing automated consent tagging, retention policies, and regulatory enforcement across protected data tiers.
- Applied DAMA-DMBOK governance standards, establishing data-quality KPIs, rule-based anomaly detection, lineage visibility, and stewardship workflows; leveraged Alation for metadata cataloging across Snowflake, Tableau, Power BI, and AWS S3, enabling governed discovery and traceability with 100% audit compliance.
- Enhanced metadata visibility and analytics readiness by integrating Reltio and Alation, enabling governed data cataloging and improving data discoverability across clinical, operational, and digital experience teams.
- Directed CAPEX planning and resource allocation for MDM modernization, aligning investments to Humana's enterprise data strategy and modernization roadmap.
- Provided technical leadership and mentorship to data engineers, governance analysts, and stewards on MDM configuration, integration engineering, and data-quality optimization using SQL, Python, PySpark, and Jupyter.

MDM Architect / International SOS (UAM Consulting)

01/2022 - 06/2023

- Worked on global MDM transformation across Provider and Customer domains spanning 100+ countries and five continents, unifying 40+ decentralized systems into a centralized SAP Master Data Governance (MDG) hub, enabling consistent, compliant, and analytics-ready master data for global health, security, and travel-assistance operations.
- Reduced data errors by 40%, cut manual maintenance effort by 25%, and improved audit pass rates by 30% by embedding SAP-MDG-based validation, lineage tracking, and privacy tagging aligned to HIPAA, GDPR, and healthcare data-protection standards.
- Designed global master-data models, entity hierarchies, and workflow automation in SAP MDG, harmonizing regional taxonomies and automating credential verification, enrichment, and validation (e.g., provider licensing, banking details, healthcare attributes). Enabled near real-time synchronization across SAP ERP, CRM, and EMR/EHR platforms.
- Architected end-to-end data integration patterns using SAP MDG replication frameworks, SAP ERP IDocs, and API-based synchronization, ensuring operational consistency, traceability, and continuity across transactional and clinical systems.

- Drove large-scale data migration and quality improvement programs using SAP MDG and SAP Data Services to cleanse, enrich, and validate legacy healthcare datasets, reducing onboarding time and improving data accuracy across regions.
- Built data-quality dashboards and SLA monitoring frameworks to track accuracy, timeliness, and completeness across Provider and Customer domains; improved SLA adherence by 20% and reduced correction turnaround time by 35% through automated workflows and steward accountability.
- Established a Global Data Governance Council and DAMA-DMBOK-aligned standards, defining stewardship roles, metadata policies, and lineage controls that improved transparency, standardization, and data reuse across business units and countries.
- Delivered a governed enterprise MDM foundation with automated exception management, end-to-end auditability, and cross-region training for data stewards and system admins on SAP MDG configuration, workflow design, and master-data maintenance best practices.

Lead MDM Data Architect / Toyota Financial (Accenture)

03/2021 - 12/2022

- Led global modernization of Toyota Financial's master-data and integration ecosystem, architecting a cloud-native, event-driven MDM platform using Semarchy xDM and Azure services to unify customer, product, and contract data across 5M+ finance accounts, 10M+ insurance contracts, and \$125B+ in assets.
- Consolidated 40+ legacy data sources in 30 days, accelerating onboarding for key domains and reducing duplication and manual remediation by 30%, while improving data scalability and propagation speed by 35% for multi-brand, cross-regional expansion (Toyota, Lexus, Mazda).
- Built real-time, API-centric master-data pipelines leveraging Azure Event Grid, Logic Apps, Data Factory, Service Bus, and Azure API Management, enabling low-latency synchronization across CRM, SAP/SAP MDG, analytics platforms, and cloud data lakes.
- Designed golden-record logic (match, merge, survivorship) and automated de-duplication pipelines within Semarchy xDM, generating end-to-end master-data lineage and improving downstream analytics/reporting speed by 40%.
- Deployed Azure Blob Storage and Data Lake for scalable, governed storage of master-data history, lineage, and audit trails—enabling long-term retention, compliance controls, and AI-ready access patterns.
- Defined reusable domain blueprints and reference cloud architectures standardizing event-driven MDM patterns across regions, reducing onboarding time ~50% for new brands, markets, and dealer networks.
- Aligned match/merge engines, stewardship models, and governance workflows in partnership with enterprise data governance teams, ensuring compliance and business-rule consistency across distributed business units.
- Collaborated with Microsoft and vendor partners on cloud provisioning, licensing optimization (Semarchy / Profisee), and performance tuning for hybrid and Azure-native deployments.
- Delivered a governed, scalable MDM foundation powering mobility-finance-as-a-service, digital retail transformation, and AI-driven customer analytics positioning TFS for next-gen mobility financing and white-label financial platforms.

Data Architect – Consumer Health Data Privacy / Accenture

01/2020 - 02/2021

- Designed data privacy and regulatory frameworks for pharmaceutical clients under CCPA and HIPAA compliance mandates.
- Conducted personal data inventory mapping, gap analysis, and risk mitigation assessments.
- Developed data classification models and access control policies for sensitive health data.
- Aligned architectural solutions with legal, compliance, and data security teams.
- Built audit-ready documentation and workflows for DSAR and consent tracking.

Data Architect – State of Texas (Child Support Unit) / Accenture

06/2014 - 05/2016

- Developed enterprise data architecture and warehouse modernization across 12+ state agencies, designing scalable, compliant, and interoperable data platforms that unified 100M+ records and supported 10,000+ statewide users for cross-agency analytics, fiscal transparency, and policy reporting.
- Increased data-model reuse by 30%, reduced data-integration timelines by 25%, and improved analytics performance by 20%, standardizing enterprise data blueprints and optimizing reporting for 3,000+ analysts and executives.

- Migrated 50+ legacy databases into modern cloud ecosystems (Azure Synapse, AWS Redshift), developed 200+ automated ETL/ELT pipelines (Azure Data Factory, Python), improving data freshness by 40%, and reduced redundant infrastructure costs by 15% via shared data services.
- Established automated profiling, data-quality checks, and lineage dashboards across 3,000+ datasets, detecting 15,000+ anomalies monthly and reducing audit prep time by 50%; achieved 35% improvement in data-quality compliance across domains.
- Defined and implemented the State of Texas Enterprise Data Governance Framework (TOGAF, DAMA-DMBOK), authoring metadata standards, stewardship roles, and access-control policies adopted by 12 agencies and 200+ data teams; increased audit-readiness by 35%.
- Formed cross-agency governance councils and working groups, improving collaboration efficiency by 40%, aligning modernization roadmaps, and delivering a statewide data-architecture foundation that enables shared analytics, modernization, and cost-efficient operations across the public sector.

Lead MDM Architect (Informatica MDM) / Ahold Food Lion (Accenture)

03/2011 - 05/2014

- Led MDM and data-product modernization for Ahold Delhaize's Food Lion retail operations (~1,000+ stores), architecting an Informatica MDM platform unifying product, supplier, location, and customer data to support omnichannel retail, supply-chain optimization, and analytics enablement.
- Reduced product and supplier data inconsistencies by 30%, shortened supplier onboarding by 40%, and accelerated product launches by 35%, improving inventory accuracy by 25% and saving 5,000+ labor hours annually across merchandising and supply-chain teams.
- Built reusable ingestion and integration pipelines using Azure Data Factory, Informatica Cloud, and Python/PySpark, processing 50M+ records monthly and ensuring real-time synchronization between ERP, PIM, POS, and CRM systems.
- Developed golden-record entity models and API-driven integrations to enable accurate, consistent data flow between retail, analytics, supply-chain, and digital commerce systems, driving real-time pricing and omnichannel product alignment.
- Established data-quality validation, enrichment rules, and governance frameworks (DAMA-DMBOK, TOGAF) with automated KPIs and scorecards tracking 25+ quality dimensions; improved audit readiness and traceability by 40% and cut issue resolution time by 35%.
- Formed an enterprise data-governance council defining stewardship roles, naming conventions, metadata standards, and catalog-based version control for 5,000+ master-data assets, ensuring governance transparency and scalability for future retail expansion.

EDI Data Architect / Independence Blue Cross

07/2004 - 03/2011

- Led enterprise EDI and data-integration modernization for claims, member, and provider domains, redesigning end-to-end X12-compliant EDI pipelines to support millions of daily transactions, high-volume claims processing, analytics enablement, and regulatory compliance.
- Reduced EDI latency by 30%, error rates by 25%, and SLA breaches by 40%, improving throughput by 20% and accelerating adjudication and payment cycles across all core health-plan systems.
- Architected enterprise-wide EDI framework using IBM Sterling Integrator, IBM MQ, and DataPower, integrating EDI flows into master-data hubs (IBM MDM / Reltio) to maintain golden records for provider and member data, increasing accuracy by 35%.
- Built event-driven, automated error-handling and SLA monitoring pipelines, reducing manual intervention by 45%, and implemented data-quality validation layers to ensure transformation accuracy and regulatory standard compliance.
- Established EDI Governance Framework and DAMA-DMBOK-aligned stewardship model, creating lineage dashboards, metadata catalogs, and KPIs to monitor completeness, timeliness, and transformation accuracy; reduced audit exceptions by 30%.
- Delivered a scalable, compliant, and analytics-ready EDI ecosystem that improved data traceability, reduced operational risk, and ensured adherence to HIPAA, PHI, and ACA standards, supporting reliable decision-making and real-time reporting across 5+ downstream systems.

EDUCATION

Doctorate: Organizational Leadership – Information Technology
University of Phoenix - Phoenix, AZ

MBA - University of Phoenix - Phoenix, AZ

MSP – University of Phoenix – Phoenix, AZ

Bachelors: Computer Information Management - Temple University - Philadelphia, PA

CERTIFICATIONS

- Certified Master Data Management Professional (CMDMP)
 - Reltio Configuration Specialist
 - Azure Fundamentals (AZ-900) | AWS Certified Cloud Practitioner
 - Azure Data Engineer Associate (DP-203)
 - Databricks Certified Data Engineer Associate
 - Segment CDP Implementation Certification (*in progress*)
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RESEARCH AND PUBLICATIONS

- ***Defining and Governing Master Data Ownership by Dr Usen Uboh***
Authored research providing guidance on how organizations should define, formalize, and operationalize master data ownership across business and IT, including data owner, steward, and custodian roles within enterprise governance models.
(Gartner Document ID: 835969)
- ***Distinguishing Master Data from Non-Master Data by Dr Usen Uboh***
Published guidance helping enterprises correctly identify what constitutes master data versus reference, transactional, and analytical data to improve MDM scope, governance clarity, and platform effectiveness.
(Gartner Document ID: 834418)
- ***MDM Client Strategy in a Changing Platform Landscape by Dr Usen Uboh***
Contributed analysis advising MDM clients on how to evaluate platform strategy, modernization paths, and vendor positioning amid market shifts and platform consolidation.
(Gartner Document ID: 836224)
- ***Cost Optimization Through Master Data Management by Dr Usen Uboh***
Authored research demonstrating how organizations can reduce technology and operational costs through MDM-driven deduplication, data pipeline consolidation, improved data quality, and governance-led architecture decisions.
(Gartner Document ID: 840060)